

COURSE SYLLABUS

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS (BSIS)

APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES

Year & Section:	4th Year – Section E
Course	APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES
Course Code	ADET413
Program	Bachelor of Science in Information Systems (BSIS)
Year & Section	4th Year – Section E
Course Type	Major / Professional Course
Suggested Term	As prescribed in the approved BSIS curriculum
Credit Units	3 Units
Prerequisite	As prescribed in the approved BSIS curriculum

I. Course Description

This course focuses on the principles and practices of developing modern software applications and evaluating emerging technologies relevant to Information Systems. Students apply requirements analysis, application design, user interface development, database integration, application programming interfaces, testing, deployment, and documentation to develop functional application solutions. The course also introduces selected emerging technologies and examines their possible applications, benefits, limitations, and responsible use in organizational and community settings.

II. Course Learning Outcomes

ILO At the end of the course, students should be able to:

- ILO 1** Explain fundamental concepts, processes, tools, and technologies used in modern application development.
- ILO 2** Analyze user and organizational requirements and translate them into appropriate application specifications.
- ILO 3** Design application components, user interfaces, database structures, and basic system architectures.
- ILO 4** Develop and integrate functional application modules using appropriate programming tools, databases, and APIs.
- ILO 5** Evaluate selected emerging technologies and determine their suitability for Information Systems applications.
- ILO 6** Conduct application testing, debugging, and refinement based on identified requirements and user feedback.
- ILO 7** Prepare appropriate technical and user documentation and present the completed application solution.

III. Weekly Topics and Syllabus Mapping

Week	Topic / Lesson	Learning Activity	Assessment
1	Introduction to Application Development	Discuss application types, development environments, and course project expectations	Diagnostic activity / participation
2	Software Development Life Cycle	Analyze planning, analysis, design, development, testing, and deployment	SDLC activity
3	Requirements Analysis	Identify stakeholders, functional and non-functional requirements	Requirements checklist
4	Application and Database Design	Prepare application flow, interface design, and database structure	Design output
5	User Interface and User Experience	Develop wireframes and improve navigation and usability	UI/UX prototype
6	Application Development Fundamentals	Implement core application features and user interactions	Development output
7	Database Integration	Connect components and implement CRUD operations	Database test
8	Midterm Application Evaluation	Demonstrate current application progress and review requirements	Midterm project evaluation
9	Application Programming Interfaces (APIs)	Integrate or consume appropriate APIs and web services	API integration output
10	Application Testing and Debugging	Conduct functional testing, identify errors, and apply corrections	Test results
11	Cloud and Modern Application Platforms	Explore deployment platforms, cloud services, and scalability	Technology activity
12	Artificial Intelligence and Emerging Technologies	Explore AI, ML, IoT, blockchain, AR/VR, and other emerging technologies	Technology evaluation
13	Emerging Technology Integration	Determine an appropriate emerging technology feature or enhancement	Integration proposal / prototype
14	Application Security and Responsible Technology Use	Apply basic security, privacy, ethical, and responsible-use considerations	Security checklist
15	Documentation and Final Application Presentation	Prepare technical/user documentation; finalize application	Documentation draft / final system check

Week	Topic / Lesson	Learning Activity	Assessment
16	Final Demonstration and Presentation	Present, demonstrate, and explain the completed application	Final presentation / demonstration

IV. Learning Resources

- Course modules and instructor-provided materials
- Application development tools, programming frameworks, and software documentation
- Database development and management resources
- API and cloud platform documentation
- Selected materials and resources on emerging technologies
- Testing, evaluation, and technical documentation templates
- Institutional and course-specific project guidelines

V. Assessment Methods

Assessment Component	Weight
Laboratory activities and development exercises	25%
Application project development and implementation	30%
Technology analysis / emerging technology activities	10%
Application testing and evaluation	15%
Documentation	10%
Final presentation and demonstration	10%
TOTAL	100%

VI. General Requirements

- Students are expected to participate actively in lectures, laboratory activities, consultations, and collaborative development tasks.
- Students must complete assigned application outputs and maintain appropriate project files and documentation.
- Students are expected to conduct required testing, debugging, refinement, and evaluation activities.
- Students must observe academic integrity and properly acknowledge third-party code, libraries, documentation, and other resources used.
- Use of AI-assisted development tools must follow instructor and institutional guidelines, and students must be able to explain and demonstrate submitted work.
- Final application outputs and supporting documents must be submitted according to the instructor's approved schedule.

VII. References

- Course modules and instructor-provided materials.
- Official programming language, framework, database, API, and cloud platform documentation.
- Software engineering, application development, and Information Systems references identified by the instructor.
- Selected academic articles and reputable materials related to emerging technologies and their application.
- Institutional guidelines and other resources identified by the instructor.

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BSIS 4th Year – Section E

I-TECH COLLEGE INC. – BAGO CITY